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INTEGRATED NLP PIPELINE: FROM REGULAR EXPRESSIONS TO MORPHOLOGICAL PROCESSING

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INTRODUCTION

- **NLP:** lets computers read, understand, and generate human language.
- Raw text is messy (typos, symbols, different word forms), so it is processed step by step.
- **NLP pipeline:** a series of stages where each output feeds the next stage.
- **Flow:** Raw Text → Regex Cleaning → Tokenization → Morphology → Understanding.
- Early stages use rules (regex); later stages use language knowledge (morphology).
- ChatGPT and Claude follow the same idea at a much larger scale.

STAGE 1: REGEX & CLEANING

- **Regular expressions:** patterns used to search, match, and replace text.
- **Uses:** remove extra spaces, URLs, symbols; find emails, phone numbers, dates.
- **Examples:** \d+ matches numbers, [A-Za-z]+ matches words.
- **Normalization:** convert to lowercase and fix spacing and punctuation.
- **Tokenization:** split text into words, e.g. "NLP is fun!" → NLP | is | fun | !
- **Limit:** fast and simple, but follows fixed rules and cannot understand meaning.

STAGE 2: MORPHOLOGY

- **Morphology:** study of word structure (prefix + root + suffix), e.g. un + happy + ness.
- **Stemming:** cuts suffixes using rules, e.g. running → run, studies → studi.
- **Lemmatization:** uses dictionary and grammar, e.g. studies → study, better → good.
- **POS tagging:** labels words as noun, verb, adjective to pick the correct root.
- **Why it matters:** groups variants (run, running, ran) into one concept.
- **Trade-off:** stemming is faster; lemmatization is more accurate.

HOW CHATGPT / CLAUDE USE NLP

- **Step 1 – Input cleaning:** handles spacing, symbols, and special characters (like regex).
- **Step 2 – Subword tokenization:** un + happi + ness (modern morphological splitting).
- **Step 3 – Embeddings:** each token is converted into numbers that capture meaning.
- **Step 4 – Transformer:** attention layers understand context between words.
- **Step 5 – Generation:** predicts the next token repeatedly to form the answer.
- **Difference:** classic pipelines use hand-written rules; LLMs learn them from data.

CONCLUSION

- An NLP pipeline turns raw, messy text into structured data a machine can understand.
- Regex handles cleaning and pattern matching; morphology handles word structure.
- Each stage prepares the text for the next, so early-stage quality affects the final result.
- ChatGPT and Claude follow the same flow: tokenize, represent, understand, generate.
- The difference is that modern AI learns the rules from data instead of hand-written rules.
- Knowing the basic pipeline makes advanced language models easier to understand.


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